Nguyễn Ngọc Thảo Ly _ 1550080153 _ CNTT4

1)

```
import java.util.Scanner;
class Nguoi {
    protected String hoTen;
    protected String ngaySinh;

    public void nhap (Scanner sc) {
        System.out.print ("Nhap ho ten : ");
        hoTen = sc.nextLine ();
        System.out.print ("Nhap ngay sinh : ");
        ngaySinh = sc.nextLine ();
    }

    public void xuat () {
        System.out.println ("Ho ten : " + hoTen);
        System.out.println ("Ngay Sinh : " + ngaySinh);
    }
}
```

2)

```
class SinhVien extends Nguoi {
    private String maSV;
    private double diemTB;

    @Override
    public void nhap (Scanner sc) {
        super.nhap(sc);
        System.out.print ("Nhap ma SV: ");
        maSV = sc.nextLine();
        System.out.print ("Nhap diem TB ");
        diemTB = Double.parseDouble (sc.nextLine());
    }
    @Override
    public void xuat() {
        super.xuat();
        System.out.println ("MaSV: " + maSV);
        System.out.println ("Diem TB: " + diemTB);
    }
    public double getDiemTB () {
        return diemTB;
    }
}
```

```
    public String getHoTen() {
        return hoTen;
    }
}

3)
import java.util.*
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Nhap so luong SV: ");
        int n = Integer.parseInt (sc.nextLine());
        SinhVien[] ds = new SinhVien [n];

        for (int i = 0; i < n; i++) {
            System.out.println ("\n SV thu " + (i+1));
            ds [i] = new SinhVien();
            ds [i].nhap (sc);
        }

        Arrays.sort (ds, new Comparator < ~~SinhVien~~
                                SinhVien > () {
            public int compare (SinhVien a, SinhVien b) {
```

```
            return a.getHoTen(), compareToIgnoreCase
                                (b.getHoTen());
        }
    });
    System.out.println("\n Danh Sach Sau Sap Xep");
    for (SinhVien sv : ds) {
        sv.xuat();
        System.out.println("    ");
    }
    double max = ds[0].getDiemTB();
    for (SinhVien sv : ds) {
        if (sv.getDiemTB() == max) {
            sv.xuat();
            System.out.println("    ");
        }
    }
    sc.close();
}
}
```

4)

```
abstract class NhanVien {
    protected String maNV, tenNV, dinhDo;
    protected double LuongCoBan;

    public void nhap() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Ma NV : ");
        maNV = sc.nextLine();
        System.out.print("Ten NV : ");
        tenNV = sc.nextLine();
        System.out.print("Dinh do : ");
        dinhDo = sc.nextLine();
        System.out.print("Luong co ban : ");
```

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```
        luongCoBan = sc.nextDouble();
        sc.nextLine();
    }
    public void xuat() {
        System.out.println("Ma: " + maNV + ", Ten: "
                + tenNV + " Dinh do: " + dinhDo +
                "Luong CB" + luongCoBan);
    }
    public abstract double tinhLuong();
}

class QuanLy extends NhanVien {
    private double phuCapChucVu;

    @Override
    public void nhap() {
        super.nhap();
        Scanner sc = new Scanner(System.in);
        System.out.print("Phu cap chuc vu: ");
        phucapChucVu = sc.nextDouble();
    }
    @Override
```

```
    public double tinhLuong() {
        return luongCoBan;
    }
}

public class MainNV
    public static void main (String [] args) {
        ArrayList <NhanVien> ds = new ArrayList <> ();
        Scanner sc = new Scanner (System.in);
        System.out.print ("Nhap so NV: ");
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            System.out.println ("\n1. Quan ly \n2.
                                Nghien cuu \n3. Phuc Vu");
            System.out.print ("Chon loai: ");
            int chon = sc.nextInt();
            sc.nextLine();
            NhanVien nv;
            if (chon == 1) nv = new Quanly();
            else if (chon == 2) nv = new NghienCuu();
            else nv = new PhucVu();
```

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```
        nv.nhap();
        ds.add(nv);
    }
    System.out.println("\n Danh sach NV: ");
    for (NhanVien nv : ds) {
        nv.xuat();
        System.out.println("Luong: " + nv.tinhLuong());
    }
}
}
```